ERROR DISTRIBUTION OF AN ARTICULATED ARM MEASUREMENT SYSTEM

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ABSTRACT

Articulated arms are often used for inspection of parts. They offer advantages of a relatively large measurement volume combined with a high degree of freedom, mobility and accuracy in the 50 μm range. Their main disadvantage is that they require time consuming manual operation. With the plethora of measurement devices on the market, including articulated arms, one often has no alternative than to rely on the OEM's claims regarding the system's accuracy. Until very recently there was no common standard for calibrating articulated arms. Even now, this standard is not widely adopted. It is therefore worthwhile to study the measurement errors of articulated arms. Errors of many devices are very seldom constant, as often quoted by OEM's, or even linear throughout the measurement volume of the device. Understanding the error distribution and how it changes through the measurement volume can help users better plan the application of the device. In this paper the error distribution of an articulated arm is determined experimentally. The errors are also modelled analytically, using linear error propagation, to better understand the experimental results, and compared to the experimental results. The experiments were done on one articulated arm and no claim regarding the generality of the results is made.

Keywords: Articulated Arm, Error Propagation, Metrology.

1 INTRODUCTION

Articulated Arm Coordinate Measurement Machines (AACMM) are often used for their flexibility and accuracy. These arms can sometimes reach areas that cannot be reached by conventional CMM's or vision based systems. In addition they often have a long work envelope compared to conventional CMM's while occupying a small floor space. They are also mobile, allowing work on the shop floor or on site. One of their main disadvantages is that they normally require manual operation for the measurement of each point.

Unfortunately, until rather recently, there was no unified standard for the calibration of these systems. Even now, the ASME standard [ASME B89.4.22-2004], is not widely adopted. Users of this equipment must therefore rely on the OEM's claims of the system's accuracy. Furthermore, the claimed accuracy is often reduced to a single value whereas the accuracy of such systems as a conventional CMM is often length dependent. A thorough investigation into the accuracy of these systems is therefore needed. In this paper it will be shown how the error varies through the measurement volume and an analytical model will be derived to predict the accuracy at any point in the envelope. This will be compared to experimental results.

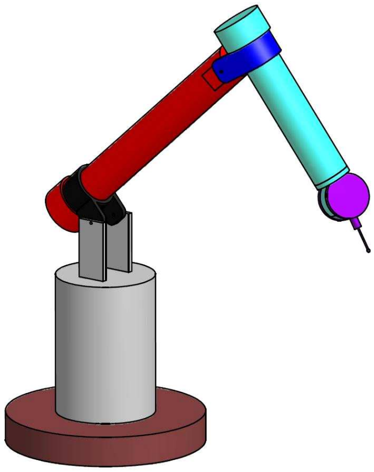

Figure 1 Schematic Representation of the AACMM used here

Conrad et al [1] have studied the problem of robotic arm accuracy through a very simplified analytical approach. They ignore the spatial dependence of the error and simply perturbed the analytical model of the arm's position with all encoders in the zero position. Their model predict an error of roughly 0.3 mm. Renaud et al [2] used a vision based system to calibrate a robot and achieved an accuracy of below 1 mm in a 1 m^3 work envelope. The vision based system gives complete pose information whereas a laser based system, which they compare their system to, only gives the position of the end effecter. Santolaria et al [3,4] did an in depth experimental investigation of an AACMM and compared their results to an analytical model that is based on the standard Denavit-Hartenberg (DH) [5] model and using a Fourier series to model the dependence on the joint angles. Their investigation also shows errors in the order of 0.3 mm.

The rest of the paper is divided in a section that describes the theory and implementation of the error propagation model that is used here. Next, the experimental tests conducted will be described and the data will be analysed. Finally, the analytical error model will be used to study the error behaviour and draw certain conclusions on the system errors after comparing the analytical model to the experimental model. The paper ends with a section on conclusions.

2 THEORY

2.1 Error propagation

The Law of Error propagation is well known [6-11] in computer vision, but since its derivation for explicit functions is simple and may be useful for readers unfamiliar with it, it is given here.

Let $F(x)$ be a vector valued function of the vector x. Then the first order Taylor expansion of the series can be written as

$$F(x) = F(\bar{x}) + J(\bar{x}) \cdot (x - \bar{x}) + \varepsilon(\|x - \bar{x}\|^2) \qquad (1)$$

$J(x)$ is the Jacobian matrix of $F(x)$ and ε is an error term of the order indicated in the brackets. The vector $\bar{x}$ is an estimation of x, in this case x could be a sample of the parameter space and then $\bar{x}$ will be the average of all the samples. If it can be assumed that the last term in the Taylor series is sufficiently small, this equation can be rewritten as

$$F(x) - F(\bar{x}) = J(\bar{x}) \cdot (x - \bar{x}) \qquad (2)$$

The covariance matrix can be written as

$$\Lambda_F = \frac{1}{n-1}\left(F(x)-F(\bar{x})\right)\cdot\left(F(x)-F(\bar{x})\right) \tag{3}$$

The dot product of two matrices, $A \cdot B$ can be replaced by AB^T. This then gives

$$\Lambda_F = \frac{1}{n-1}\left(F(x)-F(\bar{x})\right)\left(F(x)-F(\bar{x})\right)^T \tag{4}$$

The term in the brackets can now be replaced by equation (2).

$$\Lambda_F = \frac{1}{n-1}\left(J(\bar{x})\cdot(x-\bar{x})\right)\left(J(\bar{x})\cdot(x-\bar{x})\right)^T \tag{5}$$

Remember that $(AB)^T = B^T A^T$, then

$$\Lambda_F = \frac{1}{n-1} J(\bar{x})(x-\bar{x})(x-\bar{x})^T J(\bar{x})^T \tag{6}$$

Since the covariance matrix of *x* is

$$\Lambda_x = \frac{1}{n-1}(x-\bar{x})(x-\bar{x})^T \tag{7}$$

then (6) can be rewritten as

$$\Lambda_F = J(\bar{x})\Lambda_x J(\bar{x})^T \tag{8}$$

Equation (8) is very useful to determine the covariance of some process output if the input covariance is known and the Jacobian of the process function can be determined. A system like an articulated arm can be modeled mathematically, therefore the error propagation model can be used to determine the system covariance if the variances of the input sensors are known. This will be done later.

For metrology applications, it also turns out that there is a very useful way of representing the covariance graphically. Following [12], isoprobability ellipsoids will be used. The isoprobability ellipsoid represents the covariance at a point, say $\bar{x}$. The three main axes of the ellipsoid represent the eigenvectors of the covariance matrix [13] in the case that it is a 3x3 matrix, which is the case here since points in 3D space is the object of investigation. The lengths of the ellipsoid axes depend on the eigenvalues of the covariance matrix [13], where the square root of the eigenvalue gives one standard deviation. The eigenvector associated with the largest eigenvalue will be the main axis of the ellipsoid, and so forth for the smaller eigenvalues. If the measurement noise is Gaussian, then it can be said that a certain percentage of the measurement results must lie within the ellipsoid, depending on the multiple of the standard deviation that is represented by the ellipsoid. This representation will be used here.

2.2 DH-Model

The Denavit-Hartenberg (DH) model [5] was primarily developed for representing the forward kinematics of robotic arms. The model operates on the basis that a reference axis system is selected for every rotational joint and it is then related to the previous joint's axis system. By propagating through from the first reference point to the last, the end coordinate and orientation are obtained. This model is thus perfect for use with the AACMM since this represents the simplest way of interpreting the data from the encoders mounted on the arm.

The DH model requires a transformation matrix, *T*, to be generated to determine the coordinates of the final point. Since the AACMM in consideration has 6 joints the *T* matrix is computed as follows.

$$T_0^6 = A_1 A_2 A_3 A_4 A_5 A_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & X \\ r_{21} & r_{22} & r_{23} & Y \\ r_{31} & r_{32} & r_{33} & Z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

The matrix A_i is the transformation matrix of the i^{th} joint and it can be shown that it has the following form [5]

$$A_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

Here, θ_i is the rotational angle of the i^{th} encoder, α_i is the twist angle, d_i is the joint offset and a_i is the joint length as defined in [5].

3 ERROR MODELLING

The Denavit-Hartenberg parameters for the AACMM used here are given in Table 1. These parameters were obtained from the system software. They will be considered fixed and known here, although it must be stated that they cannot be known with zero uncertainty. These parameters will be used in equations (9) and (10) to determine the expected position of the probe. The schematic model of the AACMM is shown in Figure 2. The local coordinate system of each link is shown together with the encoder angle, θ_i.

Table 1 Summary of DH-Parameters

Link	a_i [mm]	α_i	d_i [mm]
1	0	90°	455
2	65	-90°	0
3	0	90°	-495.42
4	65	-90°	0
5	0	-90°	-340.82
6	117.716	180°	0

The Jacobian, needed in equation (8), will be a 3x6 matrix as follows

$$J = \begin{bmatrix} \dfrac{\partial X}{\partial \theta_1} & \cdots & \dfrac{\partial X}{\partial \theta_6} \\ \dfrac{\partial Y}{\partial \theta_1} & \cdots & \dfrac{\partial Y}{\partial \theta_6} \\ \dfrac{\partial Z}{\partial \theta_1} & \cdots & \dfrac{\partial Z}{\partial \theta_6} \end{bmatrix} \tag{11}$$

where X,Y,Z are the coordinates of the probe, obtained from the last column of T_0^6. The partial derivative, $\dfrac{\partial}{\partial \theta_i}$, is determined by taking $\dfrac{\partial A_i}{\partial \theta_i}$ and multiplying the other A_i with it in equation (8), e.g.

$$\frac{\partial T_0^6}{\partial \theta_3} = A_1 A_2 \frac{\partial A_3}{\partial \theta_3} A_4 A_5 A_6 \tag{12}$$

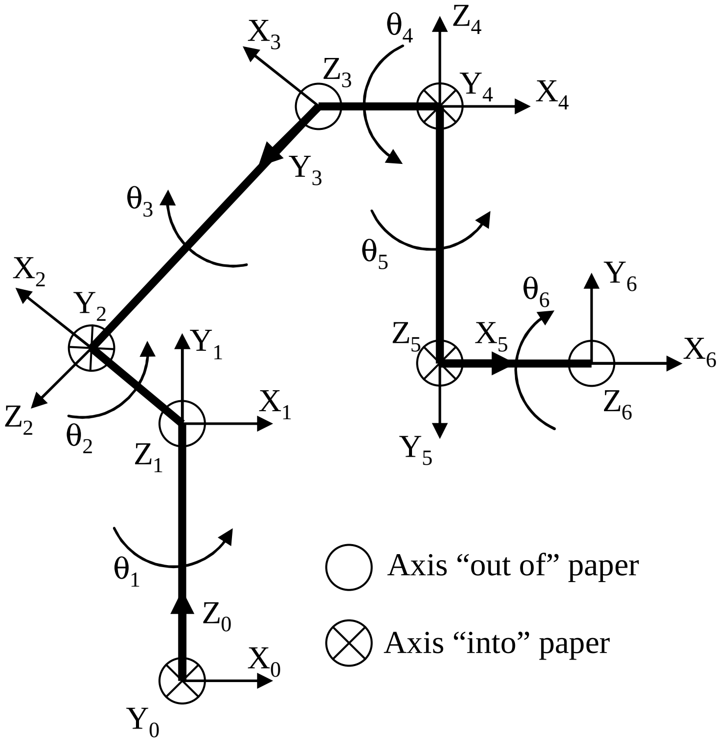

Figure 2 DH Model of the AACMM used in this Investigation (not to scale)

Note that

$$\frac{\partial A_i}{\partial \theta_i} = \begin{bmatrix} -\sin\theta_i & -\cos\alpha_i\cos\theta_i & \sin\alpha_i\cos\theta_i & -a_i\sin\theta_i \\ \cos\theta_i & -\cos\alpha_i\sin\theta_i & \sin\alpha_i\sin\theta_i & a_i\cos\theta_i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{13}$$

The variance of the AACMM encoders will be assumed to be independent of each other, therefore the covariance matrix of the encoder values will be a diagonal matrix, as follows

$$\Lambda_\theta = \begin{bmatrix} \sigma_{\theta 1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_{\theta 2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_{\theta 3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\theta 4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_{\theta 5} & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\theta 6} \end{bmatrix} \tag{14}$$

It will furthermore be assumed that $\sigma_{\theta i} = \sigma_{\theta j}$ for all *i*,*j*=1..6.

Normally, Λ_θ must be known *a priori*, however in this case it was not. The matrix will be determined *a posteriori*. Since the error law is linearly dependent on Λ_θ, it will not have an effect on the shape of the error distributions, just the scale. The shape and trends of the error distribution should therefore be the same for the analytical and experimental results. In this study only the shape and trends of the error distributions will be verified.

There can of cause also be other sources of error, e.g. manufacturing errors in the AACMM itself. These errors will also not be considered as part of this study.

4 EXPERIMENTAL VERIFICATION

4.1 Experimental Procedure

The AACMM's accuracy was experimentally verified by testing 27 positions, roughly equally spaced within a box of 550x314x533 mm. A test object, with a conical seat, was mounted on a CMM (Coordinate Measurement Machine), see Figure 3 and Figure 4. The probe was positioned in the conical seat. The conical seat provides a high degree of positioning repeatability of the probe ball. The test object was positioned with the CMM at the 27 test positions. The CMM was used, since its calibrated volumetric accuracy is about 6 µm. This is significantly more accurate than the claimed accuracy of the AACMM of 50 µm, thus providing accurate reference points. The CMM was moved to a position and then the position was measured with the AACMM by placing the probe ball within the conical seat. This measurement was repeated 35 times before the CMM was moved to the next position.
The test volume is much smaller than the measurement volume of the AACMM, but the experiments were limited by the measurement volume of the CMM.

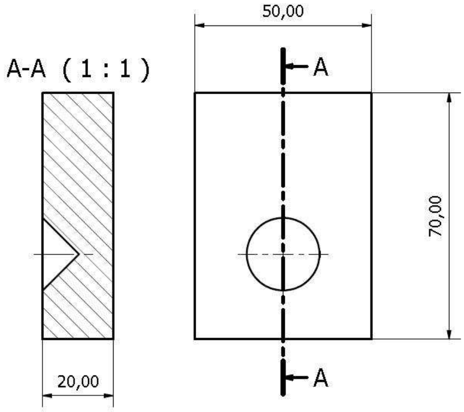

Figure 3 Test Object with Conical Seat

Figure 4 Properly Seated Probe

4.2 Data Processing

While the conical seat in the test object did contribute towards repeatable positioning of the probe, there was still a risk of measurement errors due to operator mistakes. In order to minimize the contribution of human error to the error distribution, the 3 points with the largest deviation from the CMM coordinate where removed from the dataset. This still left 32 points in the test sample. This was done for each of the 27 measurement positions.
The covariance matrix of the remaining 32 points was calculated and then the eigenvectors and eigenvalues were determined. The results were represented with error ellipsoids. This gives an indication of the repeatability of the measurements, since the covariance matrix represents the variance about the mean position of the points. The isoprobability ellipsoids are shown in Figure 6. Note that in all the isoprobability figures that the AACMM was standing at the origin of the coordinate system. Histograms of the results were also determined. A histogram is shown in Figure 5 (right) for position 22, which had the largest maximum deviation.
In order to determine the accuracy of the system, the CMM and AACMM datasets were aligned using a least squares minimization of the distance between the corresponding CMM and AACMM points. For each of the 27 positions, the deviation of each of the 32 AACMM points from the CMM position was calculated. The scaled vectors of these deviations are represented in Figure 7. A histogram of the absolute value of the deviation from the CMM position for position 22, which had the worst result, is shown in Figure 5 (left).

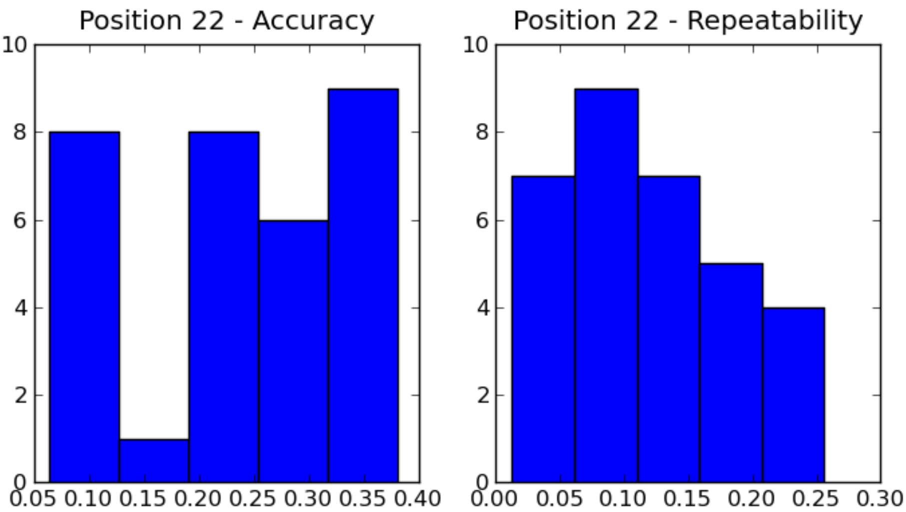

Figure 5 Histogram of Experimental Accuracy Repeatability Distribution for Position 22

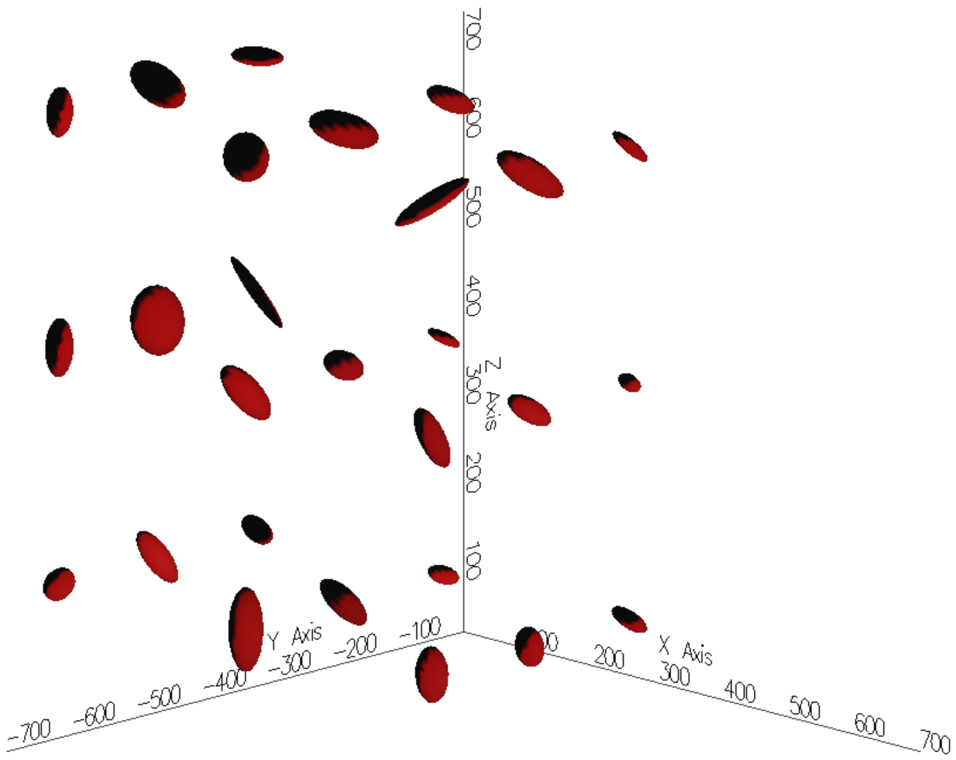

Figure 6 Isoprobability Ellipsoids Indicating Repeatability of Experimental Results Obtained from Actual Measurement taken with the AACMM

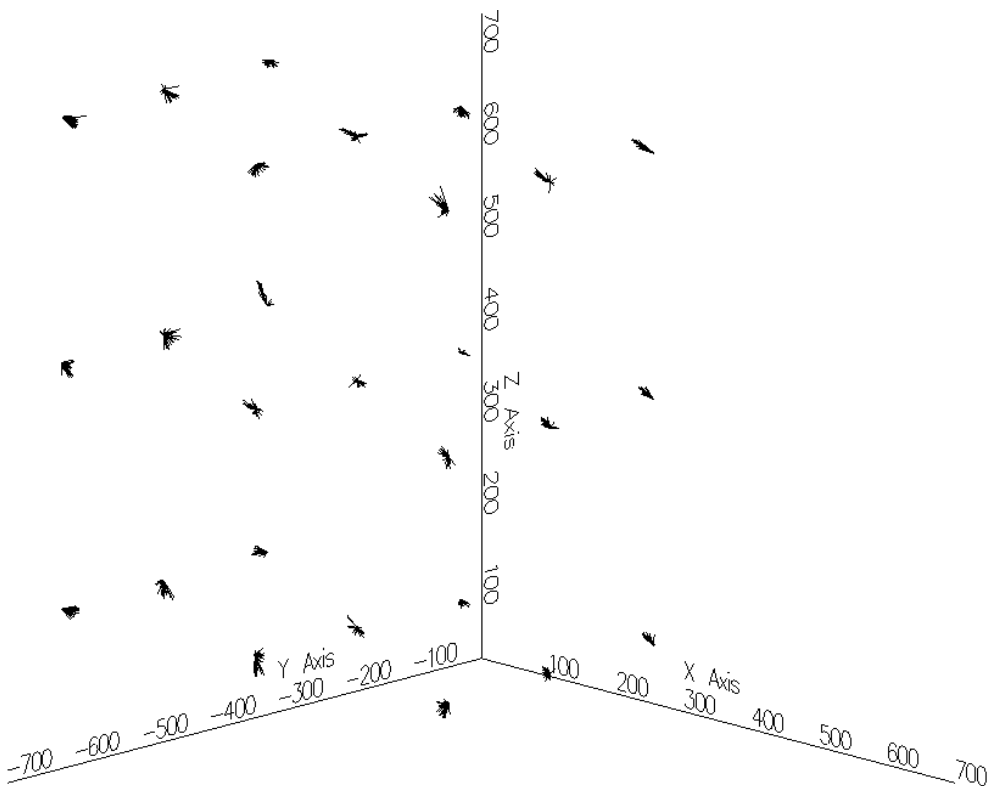

Figure 7 Scaled Vectors indicating Deviation of AACMM points from the CMM Reference Point

4.3 Discussion of Experimental Results

The range of the repeatability of the AACMM as indicated for position 22 in Figure 5 is fairly large at 0.25 mm. The rest of the 27 positions gave similar results. In fairness to the OEM it must be stated that a repetition of the experiment is needed to verify the results, but it is comparable to results from literature as mentioned earlier. Unfortunately at the time of writing this has not yet been done. The trend in the isoprobability ellipsoids of Figure 6 seems to be that as the probe moves away from the origin, where the AACMM was positioned, the deviations become larger. This is understandable, since as the arm becomes longer, it can be expected that the same error in the encoders will cause a larger probe deviation.
The comparison of the AACMM measurements with the CMM positions shows, for position 22 in Figure 5 (left), that the accuracy is slightly worse than the repeatability, at about 0.35 mm. The other points showed similar results. The error vectors in Figure 7 show that for some of these points the deviation is biased. There does not appear to be any trend in the magnitude of the deviations. The experimentally determined accuracy, although of the same magnitude as the results reported in the literature, is still much worse than the OEM's claim.

5 ANALYTICAL EVALUATION

The error model presented earlier was used to verify the experimental results. The same 27 positions were used. Constant, independent, noise was added to each encoder. The results are shown in Figure 8. The trend is more apparent for the analytical results. The error clearly grows as the arm extends. It is also interesting to note that the minor axis of the ellipsoids is orientated towards the base of the arm, as can be seen in Figure 9. However, it was found that the orientation of the ellipsoids is dependent on the orientation of the arm. Obviously, for all these 27 positions, there are infinitely many ways in which the arm can be orientated. For the purpose of this investigation, the arm's orientation was constrained so that all the links will move in a plane that is perpendicular to the XY plane. The effect of the arm's orientation on the error distribution was not part of this study.

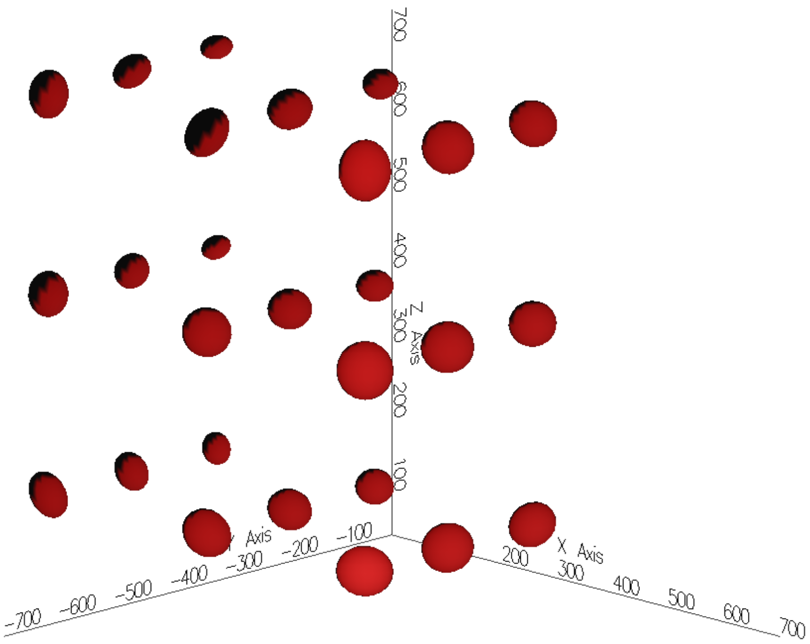

Figure 8 Scaled Isoprobability Ellipsoids obtained from Analytical Error Model.

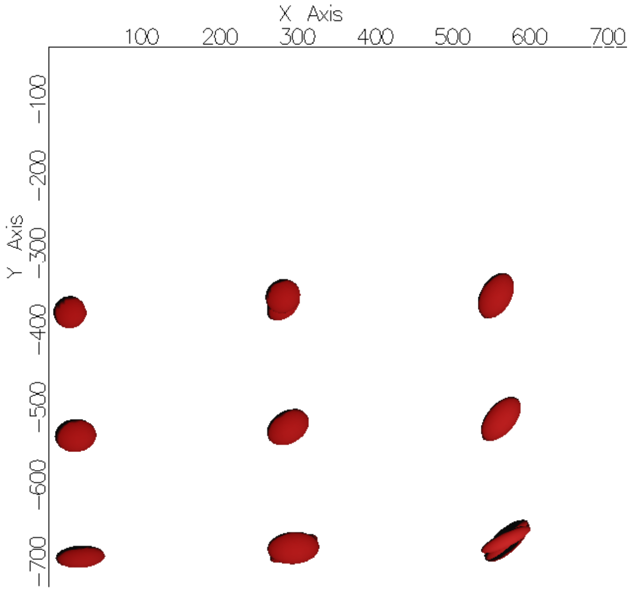

Figure 9 Top View of Scaled Isoprobability Ellipsoids

The error was also checked over a longer range than the limited range for the CMM comparison. Eight points along the X axis was tested starting at x=200 mm and going almost to the limit of the arm's range at x=900 mm. The trend of growing errors is clearly visible in Figure 10. Looking at the ellipsoids from the top, as shown in Figure 11, it is also interesting to note that the ellipsoids towards the furthest reach of the AACMM are much longer in the Z and Y directions (hereafter referred to as the tangential direction) than in the X direction (hereafter referred to as the radial direction). It appears that for this orientation of the AACMM it is more difficult to place the probe accurately in the tangential direction than in the radial direction. This result is probably more significant for robot arms of a similar configuration to the AACMM, since for a robot doing e.g. pick and place operations, it is often the former accuracy that is more important.

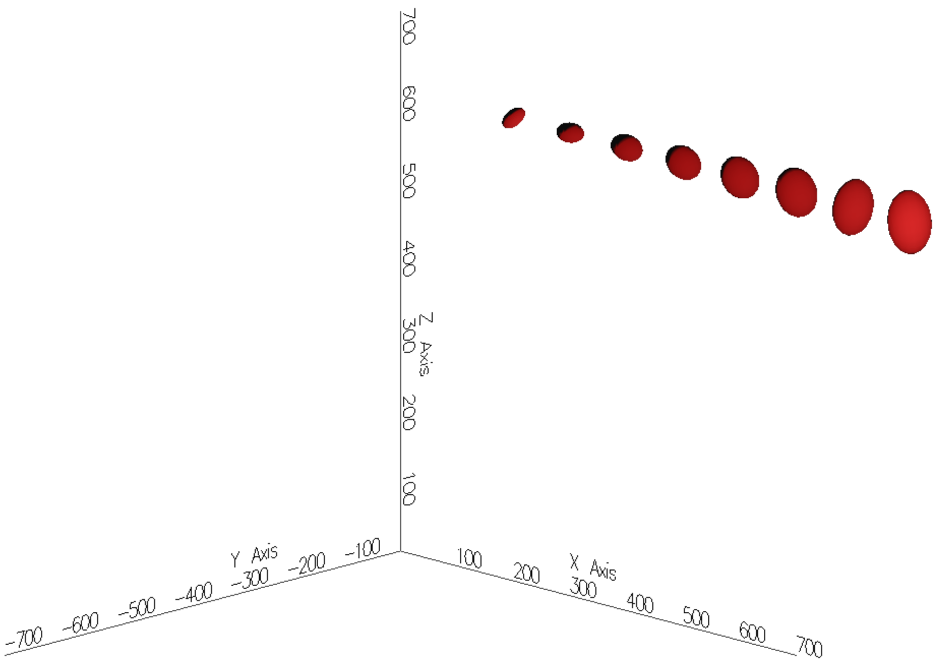

Figure 10 Scaled Isoprobability Ellipsoids along the X-axis from X=200 mm to X=900 mm

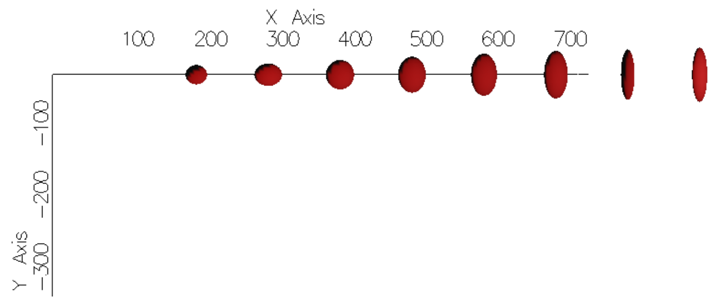

Figure 11 Scale Isoprobability Ellipsoids along the X-axis, Top View

A possible explanation for the shape of the ellipsoids in Figure 11 is that, for the selected orientation of the AACMM, there are two joints that essentially controls the radial distance (joints 2 and 4) while the other four all contribute to the tangential positioning. Since more joints contribute to the latter deviations, it seems that their combined effect could have a significant influence on the tangential error.

6 CONCLUSION

It was seen that the AACMM error distribution is dependent on the probe position and arm orientation. The latter effect was not investigated in depth, but was observed during the analysis of the experimental and analytical results. It also seems, from the error modeling, that the errors in the tangential directions are more significant than in the radial direction, for the orientation of the arm as selected in this study. This result is perhaps more significant for robot arms of a similar configuration to this AACMM. It was also seen that the trends in the analytical model of the error, while considering only the encoder errors and assuming perfect geometry, compares reasonably well with experimental results. The empirical deviations of the AACMM measurements from the "true" measurement were larger than expected.

7 ACKNOWLEDGEMENTS

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8 RECOMMENDATIONS

Further investigation into the effect of the arm's orientation on the accuracy is required. The error model proposed here is suitable for this investigation. It is also recommended to repeat the experimental tests in order to verify the results.

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